

The empirical recurrence for OEIS A229540, recovered from its tail

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8 September 2026

Abstract

OEIS A229540 records “Empirical recurrence of order 92” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 166$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 95$. Its last coefficient is $c_{92} = -358848921600 \neq 0$, so no further tail satisfies anything shorter; any order-92 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A229540 is “Number of defective 3-colorings of a 7 X n 0..2 array connected horizontally, diagonally and antidiagonally with exactly one mistake, and colors introduced in row-major 0..2 order.”. It has offset 1 and begins

0, 4212, 59626, 1379512, 26566266, 544403948, 10551803060, 205336122417, ...

The entry states:

Empirical recurrence of order 92 (see link above).

As of the “Last modified” line on the live entry (Apr 27 2021 21:14:58, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 166$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 166$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 92: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 92.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 332$ exact terms from $n = 3$ on, it returns order 92 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{92} c_i a(n-i),$$

c_1	30	c_{24}	21416120880867088671	c_{47}	−260217902457926510525842880	c_{70}
c_2	197	c_{25}	−334909194484250884666	c_{48}	−46780091502781820146550004	c_{71}
c_3	−12056	c_{26}	108926140597952574475	c_{49}	378608660099265110095165770	c_{72}
c_4	14511	c_{27}	2480251607914496352570	c_{50}	−25488165535199162300184908	c_{73}
c_5	2017390	c_{28}	−2699894245713719294427	c_{51}	−457890795112286612302601862	c_{74}
c_6	−8060915	c_{29}	−13973313428724954433622	c_{52}	122445722330547032022230107	c_{75}
c_7	−185636236	c_{30}	25811507233894136426344	c_{53}	462422204872264066264157910	c_{76}
c_8	1087320803	c_{31}	58791572543179430012732	c_{54}	−204707188451253471197070047	c_{77}
c_9	10356048186	c_{32}	−166683967316751162870151	c_{55}	−390258265889628349896329082	c_{78}
c_{10}	−81224725193	c_{33}	−171597808404116193455634	c_{56}	237882705620154513689885947	c_{79}
c_{11}	−353985149618	c_{34}	809591476013199002038665	c_{57}	274413682791241542736118964	c_{80}
c_{12}	3927882853261	c_{35}	240038482040743508716760	c_{58}	−215174848545185504960377028	c_{81}
c_{13}	6398527053780	c_{36}	−3080322493794220987920703	c_{59}	−159516491993685379361152624	c_{82}
c_{14}	−131174385822445	c_{37}	679816116805910949156418	c_{60}	157619787425584108318912012	c_{83}
c_{15}	13819473129560	c_{38}	9350556757030278859850399	c_{61}	75454851759378563566210360	c_{84}
c_{16}	3121390903368707	c_{39}	−6032503701480989231014916	c_{62}	−95129048239290626956135928	c_{85}
c_{17}	−4373525807445942	c_{40}	−22787955927149786792109372	c_{63}	−28109266117252855830230624	c_{86}
c_{18}	−53507698611290795	c_{41}	23969096082760009784044324	c_{64}	47665004363786728779994344	c_{87}
c_{19}	137910738356143036	c_{42}	44376897015817697785524725	c_{65}	7602930379685353723327064	c_{88}
c_{20}	653084611697956004	c_{43}	−67331068706595320172099462	c_{66}	−19872571306492254656444240	c_{89}
c_{21}	−2595675919524661840	c_{44}	−67556194212376912422917789	c_{67}	−1068669851347866253326272	c_{90}
c_{22}	−5332770483791331605	c_{45}	147233177947369531429288052	c_{68}	6884444285883411567950720	c_{91}
c_{23}	34260614276161093934	c_{46}	75270444092831350873578999	c_{69}	−218656428611795643258272	c_{92}

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 95$, and it is the recurrence OEIS A229540 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{92} - c_1x^{92-1} - \dots - c_{92}$. Its constant term is $-c_{92} = 358848921600$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 92 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 92, so $\deg q = 92$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 92, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 92, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -358848921600 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-92 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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